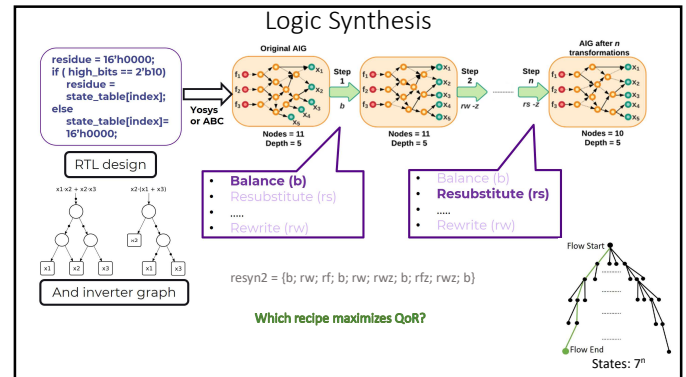


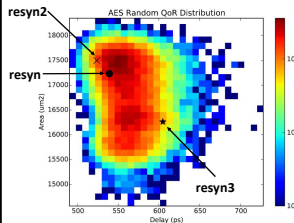
# Machine Learning for Logic Synthesis



## Which recipe?

### Experiment:

- Design: 128-bit AES core
- 6-synthesis transformations (balance, restructure, rewrite, refactor, rewrite-z, refactor-z)
- 50K recipes (Length = 24, each transformation appeared 4 times)

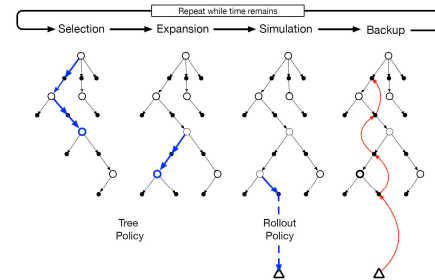


### Takeaways:

- Given the same set of synthesis transformations, QoR varies with recipes
- The search space of recipes is large
- More than 50% of the random recipes are performing better than resynX

Can ML models trained on **past data** steer search towards rewarding paths?

## Monte-Carlo Tree Search



**Tree policy:** E-greedy or Upper-Confidence-Bound (UCB) selection. **Rollout Policy:** any simply policy e.g., random

Ref: Sutton and Barto, Reinforcement Learning: An introduction, MIT Press

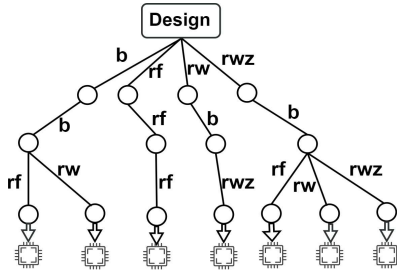
## MCTS

### Search Space

Choices: {b, rf, rw, rwz}  
Recipe length: 3

### Node information

Cumulative reward: r  
Node visit count: v



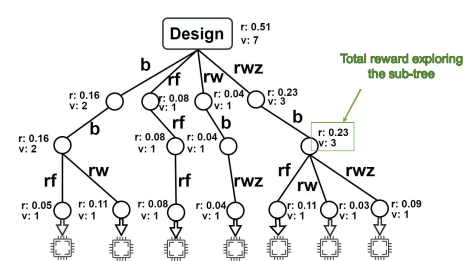
## MCTS

### Search Space

Choices: {b, rf, rw, rwz}  
Recipe length: 3

### Node information

Cumulative reward: r  
Node visit count: v



## MCTS: Selection

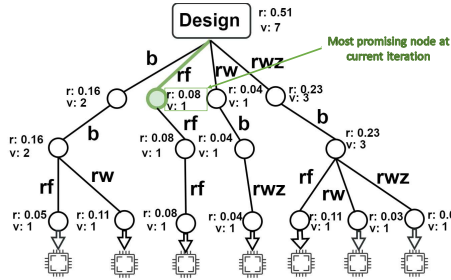
Average reward:

$$Q(n) = \frac{r(n)}{v(n)}$$

Exploration term:

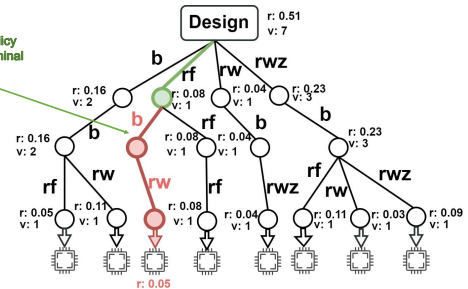
$$U(n) = \sqrt{\frac{\ln(\eta_{parent})}{v(n)}}$$

$$Score = Q(n) + U(n)$$



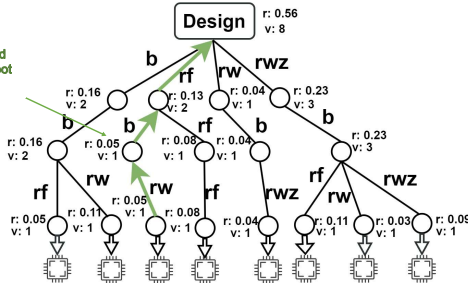
## MCTS: Expansion and Rollout

Random policy rollout till terminal

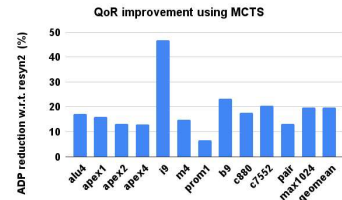


## MCTS: Backpropagation

Update v(n) and reward r(n) till root

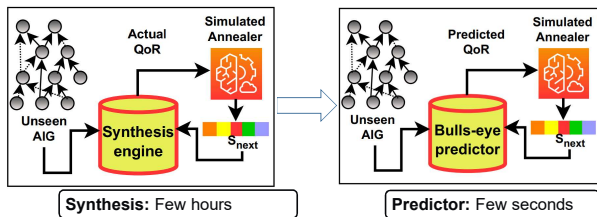


## MCTS Performance



Baseline: Area Delay Product(ADP) using resyn2

## "Bulls-eye" QoR Predictor



Synthesis: Few hours

Predictor: Few seconds

But, where is the labeled data to train the predictor?

Chowdhury et al., "Bulls-eye: Active Few-shot Learning Guided Logic Synthesis" in IEEE TCAD, 2022

## OpenABC-D: Large Scale IC Synthesis Dataset

Communication protocols

Controller

Crypto

DSP

Processors

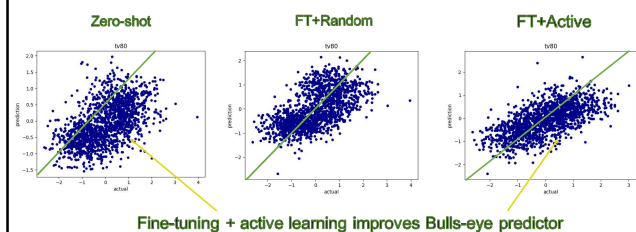
| Characteristics of Benchmarks |       |       |       |        |        |     |                                     |  |  |
|-------------------------------|-------|-------|-------|--------|--------|-----|-------------------------------------|--|--|
| IP                            | PI    | PO    | N     | E      | I      | D   | Function                            |  |  |
| ip1                           | 254   | 236   | 4239  | 8676   | 5524   | 35  | Serial peripheral interface         |  |  |
| ip2                           | 177   | 129   | 1199  | 2499   | 1198   | 15  | Balanced serial bus protocol        |  |  |
| ip3                           | 104   | 90    | 862   | 896    | 434    | 10  | Single bit PCM                      |  |  |
| ip4                           | 132   | 80    | 467   | 594    | 413    | 10  | USB 1.1                             |  |  |
| ip5                           | 125   | 125   | 813   | 1351   | 788    | 9   | Simple asynch serial controller     |  |  |
| ip6                           | 829   | 782   | 4587  | 9876   | 4768   | 29  | Wireless DMA bridge                 |  |  |
| ip7                           | 164   | 132   | 530   | 1992   | 1084   | 12  | MC68HC110 based SPI interface       |  |  |
| ip8                           | 3429  | 3137  | 15457 | 42232  | 22719  | 29  | PCI controller                      |  |  |
| ip9                           | 2122  | 2075  | 4760  | 9715   | 42138  | 24  | WISHBONE Crossbar                   |  |  |
| ip10                          | 10731 | 10422 | 47646 | 144790 | 84799  | 34  | Ethernet IP core                    |  |  |
| ac97                          | 2139  | 2137  | 1464  | 2565   | 14326  | 11  | Wireless ac97                       |  |  |
| ac97_1                        | 1197  | 963   | 4637  | 37146  | 34925  | 36  | Wireless core controller            |  |  |
| ip11                          | 1792  | 8413  | 8254  | 17441  | 10948  | 86  | Black point RISC-V processor engine |  |  |
| ip12                          | 17321 | 17063 | 10534 | 127713 | 14107  | 23  | Wireless enhanced VMA&D controller  |  |  |
| ip13                          | 303   | 64    | 8971  | 18996  | 4686   | 30  | DRS3 encryption                     |  |  |
| ip14                          | 480   | 529   | 29023 | 50739  | 26844  | 27  | AES-128 (128bit)                    |  |  |
| ip15                          | 1943  | 1942  | 17616 | 32674  | 18496  | 76  | SHA256 hash                         |  |  |
| ip16                          | 1973  | 1845  | 45468 | 93463  | 36180  | 43  | AES-128 (128bit)                    |  |  |
| ip17                          | 3087  | 2604  | 40778 | 84040  | 47391  | 42  | AES-128 (128bit)                    |  |  |
| ip18                          | 410   | 181   | 4576  | 9467   | 5986   | 47  | FIR filter                          |  |  |
| ip19                          | 494   | 441   | 8978  | 14287  | 8596   | 73  | FIR filter                          |  |  |
| ip20                          | 4942  | 4789  | 14773 | 24131  | 14600  | 40  | JPEG encoder                        |  |  |
| ip21                          | 37603 | 37419 | 24152 | 52053  | 117310 | 43  | Inverse DFT                         |  |  |
| ip22                          | 37607 | 37417 | 24154 | 52054  | 117310 | 43  | FFT design                          |  |  |
| ip23                          | 638   | 361   | 11528 | 23037  | 11653  | 54  | TV80 6-bit Microprocessor           |  |  |
| ip24                          | 4501  | 4111  | 32121 | 108011 | 14518  | 40  | SAR ADC core                        |  |  |
| ip25                          | 652   | 439   | 29622 | 59655  | 27142  | 819 | OpenSPIN T1 floating point unit     |  |  |
| ip26                          | 11042 | 10797 | 5294  | 13687  | 46767  | 47  | SoC with PicoBlaze core             |  |  |
| ip27                          | 2718  | 2275  | 1094  | 36703  | 21377  | 33  | OpenPilot SoC architecture          |  |  |

Table 1: Open source IP characteristics (uncompressed). Primary inputs (PI), Primary outputs (PO), Nodes (N), Edges (E), Inverted edges (I), Netlist Depth (D). Color code: Communication (red), processor, Controller, Crypto, DSP, Processor, Processor-control, Control-Communication

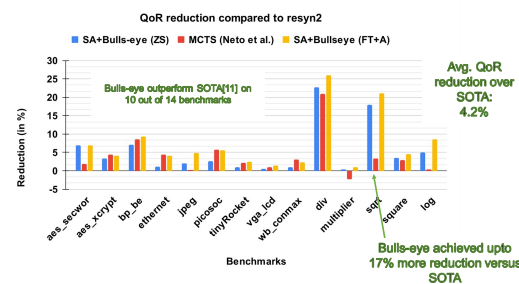
- > 29 open-source IPs
- > Open-cores, IWLs, EPFL, Github
- > Ranging from 400-250K nodes
- > Comm., control, crypto, DSP, procs



### Fine-tuning took Bulls-eye One Step Ahead



### Bulls-eye Performance on Area Reduction



### Summary

- **First work** utilizing past synthesis data + fine-tune using **active learning** for recipe generation
- **Transferability:** Past data + Smartly selected (**few-shots**) recipes = good QoR predictor for large (unseen) designs
- **Performance:** Average **13.5x** run-time speed-up and upto **30x** speed-up on large designs compared to conventional synthesis guided exploration at Iso-QoR